



Quantifying Shear-induced Margination and Adhesion of Platelets in Microvascular Blood Flow

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Abstract

Platelet margination and adhesion are two critical and closely related steps in thrombus formation. Using dissipative particle dynamics (DPD) method that seamlessly models blood cells, blood plasma, and vessel walls with functionalized surfaces, we quantify the shear-induced margination and adhesion of platelets in microvascular blood flow. The results show that the occurrence of shear-induced RBC-platelet collisions has a remarkable influence on the degree of platelet margination. We characterize the lateral motion of individual platelets by a mean square displacement analysis of platelet trajectories, and find that the wall-induced lift force and the shear-induced displacement in wall-bounded flow cause the variation in near-wall platelet distribution. We then investigate the platelet adhesive dynamics under different flow conditions, by conducting DPD simulations of blood flow in a microtube with fibrinogen-coated wall surfaces. We find that the platelet adhesion is enhanced with the increase of fibrinogen concentration level but decreased with the increase of shear rate. These results are consistent with available experimental results. In addition, we demonstrate that the adherent platelets have a negative impact on the margination dynamics of the circulating platelets, which is mainly due to the climbing effect induced by the adherent ones. Taken together, these findings provide useful insights into the platelet margination and adhesion dynamics, which may facilitate the understanding of the predominant processes governing the initial stage of thrombus formation.

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1. Introduction

Thrombosis is a serious condition where one or more blood clots form inside the blood vessels, which prevents blood from flowing normally through the circulatory system and threatens the health of patients seriously.^{1,2} Although significant

advance has been made in treatment, the prognosis of these patients is still poor,³ and thrombosis tends to recur.^{4,5} Alterations in hemostatic balance, including changes in prothrombotic and fibrinolytic coagulation factor activity,^{6,7} platelet hyperactivity,^{8–10} and endothelial dysfunction,^{10,11} remarkably increase the risk of undesirable thrombus

formation. Among them, platelets play a vital part in the process of hemostasis via initiating the blood coagulation mechanism, due to its highly efficient activation, adhesion, aggregation, and procoagulation after tissue injury.

Blood is a biological fluid that transports oxygen and nutrients to living cells and carries away carbon dioxide and other waste products. It has four main components, namely blood plasma, red blood cells (RBCs), white blood cells (WBCs), and platelets. According to the relative concentration in whole blood, the ratio of the number of RBCs, platelets, and WBCs is approximately 600: 40: 1.¹² A normal RBC is a biconcave disk-shaped cell of $8.0\ \mu\text{m}$ in diameter and $2.0\ \mu\text{m}$ in thickness (Figure 1(a)). Under physiological conditions, the RBC is remarkably deformable, which allow itself to squeeze through microcapillaries without any damage.^{13,14} Compared with the RBC, a platelet is much smaller and stiffer. In the inactivated resting form, the platelet is generally a biconvex discoid structure of $2\text{--}3\ \mu\text{m}$ in diameter (Figure 1(a)).¹⁰ When the shear-induced lift force acts on the blood cells, the lateral migration of RBCs takes place in blood flow, leading to the formation of cell-free layer (CFL) adjacent to the vessel wall where the platelets tend to concentrate (Figure 1(b)).^{15,16} Through this underlying mechanism, platelets sufficiently approach the wall trauma, build a platelet plug,

and release the thrombin that promotes the conversion of fibrinogen (FN) to the insoluble fibrin network.¹⁷ The network structure captures more cellular components into stable coagulation, providing adequate hemostasis at the bleeding site.^{18,19} Hence, the margination and adhesion of platelets play important roles in hemostasis and thrombus formation at the site of injured endothelium.

Several significant experimental studies have been undertaken to elucidate the margination and adhesion dynamics of platelets. As early as the 1980s, Reneman et al. utilized intravital microscopy to conduct experiments *in vivo* for platelet distribution in rabbit mesenteric arterioles.^{20,21} They found excess aggregation of platelets near the vascular wall. Corattiyl and Eckstein performed *in vitro* studies to investigate the influences of blood hematocrit and wall shear rate on the blood cell distribution in microtubes with diameters ranged from $50.0\ \mu\text{m}$ to $200.0\ \mu\text{m}$.²² They showed that the proportion of platelets observed near the tube wall increased as the blood hematocrit increased in the range of 10–40%. They also found that an elevated shear rate in a medium range ($80\text{--}800\ \text{s}^{-1}$) would enhance the platelet margination, whereas this effect tended to be diminished beyond such range. Rosa et al. performed a microfluidic study to investigate the margination of rigid microparticles under different shear flow

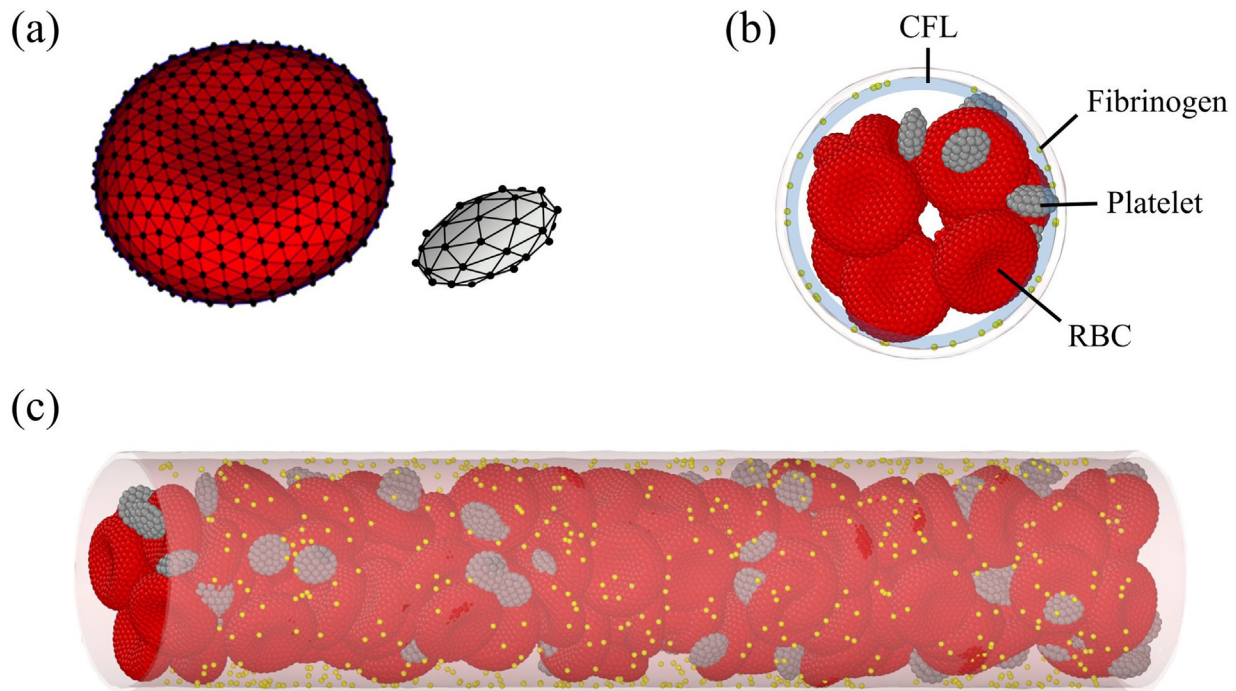


Figure 1. Schematic diagram of the blood vessel and blood cells in the DPD simulations. (a) Schematic view of the particle-based whole-cell models of RBC and platelet. (b) Front view of the blood cells inside the blood vessel. The snapshot presents the distribution of RBCs and platelets at the initial state. Fibrinogen is represented by the yellow dots on the adhesion surface of the tube wall. The cell-free layer (CFL) is displayed in light blue area. (c) Side view of the blood cells inside the blood vessel with its inner diameter $D_{in} = 20\ \mu\text{m}$.

conditions.²³ They showed that the blood hematocrit also plays a key role in the microparticle margination dynamics. Additionally, Savage et al. investigated the underlying mechanism initiating the platelet adhesion in a parallel plate flow chamber.²⁴ They pointed out that it is hard to form firm adhesion between platelet and immobilized fibrinogen under high shear rate.

Along with the aforementioned experimental studies, recent advances in computational modeling and simulations enable the investigation of a broad range of dynamic properties of blood cells in both health and disease.^{25–27} Currently, there is a great variety of modeling approaches since there is no universal solution for all blood flow related problems. Traditionally, continuum-based (or mesh-based) models,^{28–33} which treat the cell membrane and intracellular fluids as homogeneous materials, enable simulations of blood flow on macroscopic length and time scales; however, they do not provide the detailed dynamics of local sub-cellular structures. Blood cell models based on particle-based numerical methods,^{34–38} which account for the membrane structure, can resolve cellular and sub-cellular scales. Although these particle-based blood cell models have lower accuracy than the mesh-based ones, they are increasingly popular as a promising tool for modeling the microscopic blood flow in health and disease.^{25,39} For example, Yazdani et al. employed the dissipative particle dynamics (DPD) method to study the transport and dynamics of platelets flowing through microchannels with constriction.³⁷ They found that the high constriction level enhanced post-stenosis platelet aggregation, and interpreted the effects of the complex geometry on distribution of blood cells systematically. In a subsequent study, Chang et al. utilized the DPD model to examine blood cell concentration distribution and platelet margination in microtubes containing adherent WBCs.⁴⁰ They concluded that platelet margination was enhanced by increasing the flow rate or hematocrit, and found that the platelet shape and the motion of WBCs played a role in platelet distribution. Despite these interesting outcomes, many important aspects of the margination and adhesion of platelets in blood flow remain unclear, a more fundamental study on the shear-induced margination of circulating platelets should still be done. Such studies may deepen understanding of the predominant processes governing the early stages of thrombosis.

In this study, we perform a detailed computational simulation to investigate the shear-induced margination and adhesion of platelets in microvascular blood flow, with a focus on the influences of wall shear rate, fibrinogen concentration level, and adhered platelets on the margination and adhesion processes of circulating ones. The rest of the article is organized as follows: firstly, we describe the simulation methods; subsequently, we present and discuss

the simulation results; finally, we summarize the main findings and draw the conclusions.

2. Simulation Methods

2.1. Dissipative particle dynamics method

The dissipative particle dynamics (DPD) is a particle-based mesoscopic approach that describes a set of particles moving together in a Lagrangian fashion.⁴¹ In DPD simulation, each particle is regarded as a coarse-grained virtual cluster, and its position and momentum is updated in a continuous phase at discrete time steps based on Newton's second law of motion,^{42,43}

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \quad (1)$$

$$\frac{d\mathbf{v}_i}{dt} = \mathbf{F}_i, \quad (2)$$

where the mass of each particle is set to unit 1, the position $\mathbf{r}_i$ and velocity $\mathbf{v}_i$ of particle i are iterated in discrete time steps.

The resultant force $\mathbf{F}_i$ consists of three pairwise force components: conservative force $\mathbf{F}_{ij}^C$, dissipative force $\mathbf{F}_{ij}^D$, and random force $\mathbf{F}_{ij}^R$, that is,

$$\mathbf{F}_i = \sum_{j \neq i} (\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R). \quad (3)$$

The standard forms of these three pairwise interactions between particle i and particle j are represented by

$$\mathbf{F}_{ij}^C = a_{ij}(1 - r_{ij}/r_c)\mathbf{e}_{ij}, \quad (4)$$

$$\mathbf{F}_{ij}^D = \gamma_{ij}\omega_d(r_{ij})(\mathbf{e}_{ij} \cdot \mathbf{v}_{ij})\mathbf{e}_{ij}, \quad (5)$$

$$\mathbf{F}_{ij}^R = \sigma_{ij}\omega_r(r_{ij})(\zeta_{ij}/\sqrt{\Delta t})\mathbf{e}_{ij}, \quad (6)$$

where a_{ij} , γ_{ij} , σ_{ij} represent the coefficients for the conservative, dissipative, and random forces, respectively, r_c is the cutoff radius, r_{ij} is the length of the displacement vector $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$, $\mathbf{e}_{ij} = \mathbf{r}_{ij}/r_{ij}$, $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the difference in velocity between particles i and j , ζ_{ij} is a random number obeying a Gaussian distribution with zero mean and unit variance, and Δt is the time step size. Coupled through the fluctuation–dissipation theorem, the parameters γ_{ij} and σ_{ij} need to satisfy the relation $\sigma_{ij}^2 = 2\gamma_{ij}k_B T$, where k_B is the Boltzmann constant and T is the system temperature. The weight functions $\omega_r(r_{ij})$ and $\omega_d(r_{ij})$ are given by

$$\omega_r(r_{ij}) = \begin{cases} (1 - r_{ij}/r_c)^k, & r_{ij} < r_c \\ 0, & r_{ij} \geq r_c \end{cases}, \quad (7)$$

$$\omega_d(r_{ij}) = \omega_r^2(r_{ij}), \quad (8)$$

where the value of k is used to regulate the fluid viscosity. We set $k = 0.2$ in the present work.

In the DPD simulation of blood flow, it can seamlessly model the blood cells, blood plasma, and vessel walls with functionalized surfaces. As a

result, the DPD has been proven to be a powerful tool for mesoscopic modeling of microvascular blood flow.^{44–48}

2.2. Mesoscopic models of RBC and platelet at the whole-cell level

In the DPD-based cell model, the membrane of the blood cell is depicted as a two-dimensional triangular network with N_v vertices, which are connected by N_s viscoelastic springs combined with bending energy and constraints for surface-area and volume conservation.^{34,35} The total energy of the model system includes the elastic energy, bending energy, constraints of the fixed surface area and enclosed volume, which is described by

$$U = U_e + U_b + U_a + U_v. \quad (9)$$

where U_e represents the stored elastic energy associated with the spectrin network, given by,

$$U_e = \sum_{i \in 1, \dots, N_s} \left(\frac{k_B T l_{\max}}{4p} \frac{3x_i^2 - 2x_i^3}{1 - x_i} + \frac{k_p}{l_i} \right), \quad (10)$$

where l_i is the length of bond i , $l_{\max}$ is the maximum length, $x_i = l_i/l_{\max}$, p is the persistence length, and k_p reflects the strength of repulsive force. The shear modulus μ_s of cell membrane can be calculated as,^{34,35}

$$\mu_s = \frac{\sqrt{3}k_B T}{4pl_{\max}} \left(\frac{x_0}{2(1 - x_0)^3} - \frac{1}{4(1 - x_0)^2} + \frac{1}{4} \right) + \frac{\sqrt{3}k_p}{2l_0^2}, \quad (11)$$

where l_0 is the equilibrium bond length and $x_0 = l_0/l_{\max}$.

The bending energy U_b of cell membrane is modeled by

$$U_b = \sum_{j \in 1, \dots, N_s} k_b [1 - \cos(\theta_j - \theta_0)], \quad (12)$$

where k_b is the bending constant, θ_j and θ_0 are the instantaneous and spontaneous angles between two adjacent triangles with a shared common edge j .

In addition, the surface area and volume are controlled to mimic the incompressible intracellular cytosol and the area-preserving cell membrane. The corresponding energy is given by,

$$U_a = \frac{k_A(A - A_t)^2}{2A_t} + \sum_{j \in 1, \dots, N_t} \frac{k_a(A_j - A_0)^2}{2A_0}, \quad (13)$$

$$U_v = \frac{k_v(V - V_t)^2}{2V_t}, \quad (14)$$

where N_t is the number of membrane network triangles, k_A and k_a are the global and local area constraint coefficients, respectively, k_v is the volume constraint coefficients, A_j is the instantaneous area of triangle j , A_0 is the triangle area value at equilibrium, A and V are the cell membrane area and volume, and A_t and V_t are the targeted total cell area and volume, respectively.

The cell membrane interacts with the fluid and wall particles through DPD forces, and the system temperature is maintained by the DPD thermostat. To obtain the initial shape of the RBC, the vertices

of the 2D triangular network are uniformly distributed over the cell membrane surface based on the following equation,

$$z_{\text{RBC}} = \pm D_0 \times \sqrt{1 - \frac{4(x^2 + y^2)}{D_0^2}} \left[a_0 + a_1 \frac{(x^2 + y^2)}{D_0^2} + a_2 \frac{(x^2 + y^2)^2}{D_0^4} \right], \quad (15)$$

where $D_0 = 7.82 \mu\text{m}$ is the RBC diameter, $(a_0, a_1, a_2) = (0.052, 2.003, -4.491)$. The stress-free state is then obtained by performing the free energy relaxation while restricting the movement of the vertices on the cell membrane.³⁴ In addition, the platelet is modeled as an oblate-spheroid shape characterized by its aspect ratio (AR)

$$AR = a_{\min}/a_{\max}, \quad (16)$$

where $a_{\min}$ and $a_{\max}$ are the length of semi-minor axis and semi-major axis. Compared to the modeled RBC, it is considered to be much stiffer.

2.3. Platelet adhesive interactions

We employ a particle-based stochastic bond formation/dissociation model^{49–51} to capture the adhesive interactions between platelet and FN-coated adhesion surface of the tube wall. Specifically, we set a set of frozen DPD particles (yellow dots shown in Figure 1(b) and (c)) to model the adhesive ligands, which can bind to the receptors on the platelet membrane and form transient adhesive bonds within an effective distance d_f with a probability

$$P_f = 1 - e^{-k_f \Delta t}. \quad (17)$$

Meanwhile, the existing adhesive bonds can be ruptured within a critical length d_r with a probability

$$P_r = 1 - e^{-k_r \Delta t}, \quad (18)$$

where k_f and k_r characterize the bond formation and dissociation rates determined by

$$k_f = k_{f0} \exp \left(-\frac{\sigma_f(l - l_0)^2}{2k_B T} \right), \quad (19)$$

$$k_r = k_{r0} \exp \left(\frac{\sigma_r(l - l_0)^2}{2k_B T} \right), \quad (20)$$

where σ_f and σ_r represent the strengths of the effective bond formation and rupture, respectively. For existing adhesive bonds, the adhesive force between platelet and FN-coated adhesion surface is given by

$$F_s(l) = 2k_s(l - l_{s0}), \quad (21)$$

where k_s is the stiffness coefficient of the spring, and l_{s0} is the equilibrium length. The stochastic adhesion dynamics model has been successfully applied to investigate the adhesive dynamics of all three types of blood cells^{51,52,36,53} as well as circulating tumor cells.^{38,54} For a detailed description of the cell adhesion model, we refer to Refs.^{51,36}.

2.4. Model and physical units scaling

In a DPD approach, it is convenient to use reduced units; hence, it is of great importance to obtain the relationship between the DPD model units and physical units. Several recent studies have attempted to map the model units into dimensional units.^{34,35,55} A mapping strategy developed by Pivkin et al.³⁴ and further extended by Fedosov et al.³⁵ is adopted to provide an estimate of the physical length and time scales in the DPD simulations of blood flow. Following their mapping strategy, we obtain the corresponding scales,

$$[L] = \frac{D_0^P}{D_0^M}, \quad (22)$$

$$[t] = [L] \frac{\eta^P \mu_s^M}{\eta^M \mu_s^P}, \quad (23)$$

$$[F] = [L] \frac{\mu_s^P}{\mu_s^M}, \quad (24)$$

where the superscripts M and P represent the DPD model unit and physical unit, respectively. D_0 is the RBC diameter, η is the plasma viscosity, μ_s is the shear modulus of RBC membrane. In the current study, we obtain $[L] = 1.0 \times 10^{-6}$ m, $[t] = 1.8 \times 10^{-4}$ s, and $[F] = 4.7 \times 10^{-14}$ N. For details of the dimensional calibration, we refer to Refs.^{34,35,55}.

2.5. Parameter estimation

Parameters related to the membrane properties of RBC and platelet models, including the shear modulus, bending coefficient, cell surface area and volume, are summarized in Table 1. A normal RBC has remarkable deformability under physiological condition. In a pathological state, the RBC shape changes from the normal biconcave shape to a near-oblate shape, and the deformability is also weakened. Based on an analytic theory of the hexagonal network, the modeled cell membrane properties can be uniquely related to the experimentally established cell macroscopic properties (such as shear and bending moduli). Specifically, the bending constant k_b is related to the bending rigidity k_c of cell membrane via the relationship $k_b = 2k_c/\sqrt{3}$.³⁵ As suggested by Chang et al.,⁴⁰ the shear modulus of the modeled RBC is set at $\mu_{s,R} = 9.4 \times 10^{-6}$ N/m, which falls within the range of available experimental data (4– μ N/m);^{56,57} the

bending rigidity of the modeled RBC is set at $k_{c,R} = 2.4 \times 10^{-19}$ J, which falls within the range of 1.0×10^{-19} J and 7.0×10^{-19} J measured in experiments.^{58,59} In addition, the cell surface area $A_{t,R}$ is set to $132.9 \mu\text{m}^2$ and cell volume $V_{t,R}$ is equal to $127.5 \mu\text{m}^3$. The compression modulus $K = 2\mu_0 + k_a + k_d$. Generally, for a nearly incompressible sheet of springs, the area constraint coefficients should be large enough such that $k_a + k_d \gg \mu_0$. In this study, we used the values $\mu_0 = 100$ (in DPD unit) and $k_a + k_d = 40000$ for the DPD-based RBC model, which provide a nearly incompressible RBC membrane. Similarly, we set $k_v = 40000$ for the RBC volume conservation constraint. The DPD-based RBC models have been demonstrated to be able to reproduce realistic RBC mechanical, rheological and dynamic properties in both static and flow conditions.^{34–36,48}

The DPD-based platelet models are less accurate when compared to the DPD-based RBC models. Hence, they are being constantly refined to better capture the structural, mechanical, and adhesive properties of platelets.^{40,53} In this study, we consider the following several aspects to better describe the platelets:

- **Platelet shape.** For a typical platelet, its aspect ratio (AR) is ranged from 0.25 to 0.5.⁶⁰ Here, we consider a platelet to be inactivate state with an oblate-spheroid shape, and we choose $AR = 0.38$ with the length of its semi-major axis $a_{max} = 1.65 \mu\text{m}$. In addition, we model a platelet with $N_{v,P} = 48$ with its volume $V_{t,P} = 6.0 \mu\text{m}^3$ and surface area $A_{t,P} = 19.6 \mu\text{m}^2$.
- **Platelet stiffness.** Compared with healthy RBCs, a platelet exhibits larger cell rigidity; hence, we use values one order of magnitude higher for their shear modulus and bending rigidity than RBCs. Specifically, we set shear modulus $\mu_{s,P} = 4.7 \times 10^{-5}$ N/m ($= 5\mu_{s,R}$) and bending rigidity $k_{c,P} = 4.1 \times 10^{-18}$ J ($\approx 17.1k_{c,R}$).
- **Platelet adhesion.** To represent the adhesive interaction between the platelets and the endothelial cell, we coated the vascular wall with static DPD particles to model the adhesive ligands (such as fibrinogen), which can potentially interact with the receptors on the platelet membrane and form into adhesive bonds. Here, we employ the stochastic kinetic adhesion model to represent the transient bond formation and dissociation, which has been successfully applied to investigate the adhesive dynamics of all three types of blood cells^{51,52,36,53} as well as circulating tumor cells.^{38,54} As suggested by these previous simulation

Table 1 **Selected cell membrane parameters in the modeling of RBC and platelet.** N_v is the vertices number of triangular network on the membrane, l_m/l_0 is the ratio of maximum bond extension to equilibrium bond length, k_b is the bending coefficient, μ_s is the membrane shear modulus, A_t and V_t are the targeted total cell area and volume, respectively, $k_A + k_a$ is the area constraint constant, and k_v is the volume constraint constant.

Cell	N_v	l_m/l_0	k_b	μ_s	A_t	V_t	$k_A + k_a$	k_v
RBC	500	1.8	6.03	200.0	132.87	127.46	5000	5000
Platelet	48	1.8	100.00	10000.0	19.63	6.00	5000	10000

studies, we set the model parameters related to the adhesive dynamics between platelets and FN-coated surface, which are provided in Table 2.

We quantify shear-induced margination and adhesion of platelet in the blood flow. The plasma is represented by DPD fluid particles with viscosity $\eta = 1.25 \times 10^{-3}$ Pa.s. The fluid-cell interaction is achieved by viscous friction. Following Yazdani et al.³⁷ and Chang et al.,⁴⁰ parameters related to the interaction between two DPD particles, including RBC (denoted by R), platelet (denoted by P), solvent (denoted by S), and wall (denoted by W), and FN-coated adhesion surface (denoted by F) particles, are summarized in Table 3. It is worth noting that we use a relatively strong repulsive interactions between the blood cells to avoid their overlapping. Specifically, we set the coefficient $a_{ij} = 10.0$. At such a strong repulsion, we find that the overlapped blood cells gradually separate each other as the blood flow developed. We counted and found that only less than 5% of platelets are still overlapped with other blood cells after the blood flow is fully developed.

As suggested by previous simulations,^{52,36} we apply a Lennard-Jones repulsive interaction to blood cells to quantify the effect of cell overlapping on the simulation results. We run a set of new simulations and compared the platelet margination results with those in the absence of the Lennard-Jones repulsion cases at a fixed shear rate ($\dot{\gamma}_w = 880 \text{ s}^{-1}$), see Figure S1(a). It shows that the platelet counts in the CFL region for the simulation cases with the Lennard-Jones repulsive interaction is only 2.4% higher than those without considering the Lennard-Jones repulsion cases. Based on these results, we confirm that the overlapping between the blood cells had little impact on the simulation results.

With regards to the effect of cell-wall repulsion on the CFL thickness, we estimate the CFL thickness at two different a_{ij} values (Figure S1(b)). We find that the values of the CFL thickness has little changes, which is consistent with previous experimental observation by Kim et al.⁶¹ Additionally, according to our previous simulation study,

Table 2 Simulation (in DPD units) and physical (in SI units) parameters for adhesive dynamics of platelets. k_s is spring constant, l_{s0} is the equilibrium spring length, σ_f and σ_r represent the effective bond formation and rupture strengths, respectively, k_{f0} is unloaded bond formation rate, and k_{r0} is unloaded bond rupture rate.

Parameter	Simulation	Physical
Spring constant k_s	100.0	4.7×10^{-6} N/m
Equilibrium spring length l_{s0}	0.0	0.0 m
On strength $\sigma_f/k_B T$	10.58	$10.58 \mu\text{m}^{-2}$
Off strength $\sigma_r/k_B T$	3.49	$3.49 \mu\text{m}^{-2}$
Bond formation rate k_{f0}	100	$5.6 \times 10^5 \text{ s}^{-1}$
Bond rupture rate k_{r0}	0.01	56 s^{-1}

Table 3 Selected parameters used in DPD simulations. a_{ij} , γ_{ij} and σ_{ij} represent the conservative, dissipative and random coefficients, respectively, r_c is the cutoff radius. Note that R, P, S, W, F denote RBC, platelet, solvent (plasma), vessel wall, FN-coated adhesion surface, respectively.

Type	a_{ij}	γ_{ij}	σ_{ij}	r_c
R-R/R-P	10.0	10.0	1.414	1.0
R-S	0.0	45.0	3.000	1.0
R-W/R-F/P-W/P-F/S-W/S-F	2.5	30.0	2.377	1.0
P-P	2.5	10.0	1.414	1.0
P-S	0.0	10.0	1.414	1.0
S-S	5.0	20.0	2.000	1.58
W-W/W-F/F-F	0.1	30.0	2.377	1.0

the thickness of CFL, to a large extent, is mainly determined by blood hematocrit H_{CT} , where the values of the CFL thickness are estimated to be 4.6, 2.8, 2.2, and $0.8 \mu\text{m}$ at $H_{CT} = 10\%, 20\%, 30\%, 45\%$, respectively.⁴⁸ In the current study, we set $H_{CT} = 40\%$, therefore, it is reasonable to have the value of CFL thickness at about $1.5 \mu\text{m}$.

2.6. Simulation setup

Based on physical geometric considerations and previous simulation studies on blood cells in microvessels, we constructed a three-dimensional circular microtube for a microvascular model with a length of $100 \mu\text{m}$ and a diameter of $20 \mu\text{m}$ (the inner diameter $D_{in} = 20 \mu\text{m}$ and the outer diameter $D_{out} = 22 \mu\text{m}$), as shown in Figure 1(c). In this study, the blood hematocrit is set as $H_{CT} = 40\%$, and the platelet number in the blood flow is 100. In addition to the RBCs and platelets, there are also fluid particles representing the blood plasma in the microtube. In the beginning, all blood cells and blood plasma are placed randomly in the microtube. To drive fluid flow, a constant body force f is imposed on all fluid particles in the positive x direction, causing a constant uniform gradient of pressure drop $\frac{\Delta P}{\Delta x} = \rho \cdot f$, where ΔP is the pressure drop on length Δx .⁴⁸ The resulting blood flow can drive the RBCs and platelets to move along the axial direction. In the simulation, the physical time to reach a steady state is about 0.15 s, where a fully developed plug-like velocity profile is formed caused by the nonuniform distribution of RBCs. Furthermore, different pressure drops lead to diverse flow velocity profiles. The magnitude of the wall shear rate $\dot{\gamma}_w$ is approximately linearly related to the gradient of pressure drop $\frac{\Delta P}{\Delta x}$, which varies from 250 s^{-1} to 1540 s^{-1} in our simulations.

3. Results and Discussion

In this study, we focus on platelet margination and adhesion dynamics in microvascular blood flow. We first investigate the effect of wall shear rate on the

lateral transport and distribution of the circulating platelets. We then perform a detailed analysis of the impact of wall shear rate on platelet adhesion and compare the simulation results with available experimental studies. Next, by introducing the FN-coated adhesion surface within the microtube, we probe the influence of adhered platelets on the flow dynamics and transport of the circulating ones. According to the study by Freund et al.,^{62,31} when the wall shear rate $\dot{\gamma}_w$ is higher than 100 s^{-1} , the CFL region is almost independent of the flow rate; thus, the change of the CFL thickness is not considered in the current simulations.

3.1. Effect of wall shear rate on platelet margination in blood flow

The lateral migration of RBCs leads to the formation of CFL adjacent to the blood vessel wall. Reduced local viscosity in this layer would help reducing the resistance to flow in small blood vessels, which is critical for blood flow in microcirculation. Moreover, the RBCs tend to push out less deformable particles from the central region into the CFL region, leading to the margination of platelets and WBCs to the blood vessel wall, which is essential to their functions. Hence, the presence of the RBCs and the shear-induced collisions between the RBCs and the platelets have remarkable influences on the margination of platelets in blood flow. In this study, all platelets are randomly placed in the microtube at $t = 0 \text{ s}$. We then apply an externally body force to the fluid particles, and the resulting fluid flow field will affect and drive them to flow within the microtube. We plot the radial trajectories of the

circulating platelets with respect to time, see Figure 2(a). Here, we use the bold colored lines to highlight the typical trajectories of four representative platelets, which are placed at the initial position located at $r = 6.0 \mu\text{m}$ from the tube center. The blue, orange, and red trajectories represent the platelets at the wall shear rate of $\dot{\gamma}_w = 250 \text{ s}^{-1}$, 880 s^{-1} , and 1540 s^{-1} , respectively. We note that the collision frequency among the blood cells increases with increasing the shear rate, leading to strong lateral movement of platelets and fast margination to the blood vessel wall at low and moderate values of wall shear rate. After that, with the further increase of the wall shear rate, there is a reduction in the platelet margination, due to the dramatic platelet velocity fluctuations. This makes it prone for platelets to eject from the CFL region and reenter the core region. In Figure 2(a), the platelets move steadily with the fluid, and their lateral location is almost constant at $\dot{\gamma}_w = 250 \text{ s}^{-1}$. However, for $\dot{\gamma}_w = 1540 \text{ s}^{-1}$, the lateral motion of the platelets fluctuates violently, and an evident “waterfall phenomenon” is observed, which describes the abrupt margination when the platelets approach the RBC core.⁶³

To characterize the lateral transport ability of the circulating platelets, we utilize the mean square displacement $\text{MSD} = \langle |r(t) - r(0)|^2 \rangle$, where $\langle \rangle$ represents the average of all platelets in the microtube. It shows that the MSD is strongly associated with the transient lateral location of the circulating platelets, which is directly affected by the shear-induced velocity fluctuations. We analyze the alteration of the MSD at different shear rates. As depicted in Figure 2(b), the value

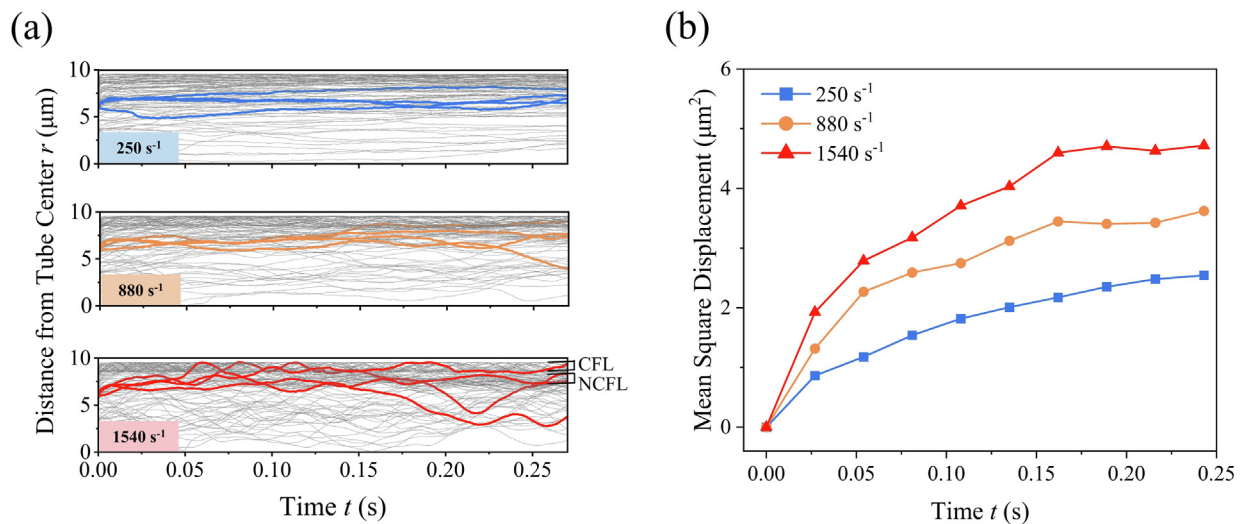


Figure 2. Lateral migration of circulating platelets in blood flow at different wall shear rates. (a) Trajectories of the circulating platelets along the radial direction at wall shear rate $\dot{\gamma}_w = 250 \text{ s}^{-1}$, 880 s^{-1} , and 1540 s^{-1} , respectively. The trajectories of four representative platelets with their initial positions located at around $r = 6 \mu\text{m}$ are highlighted by bold color lines. (b) Time-dependent mean square displacement of the center-of-mass of circulating platelets at three different wall shear rates.

of MSD increases with time and reaches a maximum value at about 0.2 s. Moreover, the magnitude of the MSD value is positively related to the shear rate, which mounts up by about 85% as $\dot{\gamma}_w$ increases from 250 s^{-1} to 1540 s^{-1} at 0.24 s.

More interestingly, we observe a stratified phenomenon of platelet distribution in Figure 2(a) at $\dot{\gamma}_w = 1540 \text{ s}^{-1}$. The nearest layer close to the wall is the CFL, and the layer next to the CFL is called the near CFL (NCFL). From the platelet trajectory diagrams, we find that the spatial separation between these two layers is approximately $0.2 \mu\text{m}$. The range of the CFL is $8.5 \mu\text{m} < r < 10.0 \mu\text{m}$, whereas the range of the NCFL is $7.0 \mu\text{m} < r < 8.3 \mu\text{m}$. The increase in wall shear rate is advantageous to the aggregation of platelets into the NCFL instead of the CFL. Figure 3(a) counts the ratios of platelets in the CFL and the NCFL at different wall shear rates at $t = 0.27 \text{ s}$, respectively. The total ratio of the circulating platelets in these two regions is almost maintained at 60–70%, and the rest of the circulating platelets enter the RBC enrichment area. With the rise of the shear rate, the count of platelets in the CFL (R_C) first increases, but then decreases rapidly. The variation tendency of the platelet ratio in the NCFL (R_N) is exactly the opposite of that in the CFL. The ratio $R_C : R_N$ reaches the maximum value of 1.95 at $\dot{\gamma}_w = 550 \text{ s}^{-1}$ and the minimum value of 0.68 at $\dot{\gamma}_w = 1540 \text{ s}^{-1}$. These simulation results reflect a competitive relationship between the RBC-platelet collisions leading to platelet margination and the shear-induced lift force driving the circulating platelets to move away from the blood vessel wall.

To show the lateral distribution of the circulating platelets in the microtube under different shear

rates, we plot the platelet concentration profiles at $\dot{\gamma}_w = 250 \text{ s}^{-1}$ (blue histogram) and 1540 s^{-1} (red histogram) at $t = 0.27 \text{ s}$, see Figure 3(b). It is clearly seen that the platelets hardly approach the center zone. In addition, at $\dot{\gamma}_w = 1540 \text{ s}^{-1}$, platelets are mainly distributed in the range of $7.0\text{--}8.0 \mu\text{m}$, with a maximum value of at around $7.5 \mu\text{m}$ from the tube center; however, at $\dot{\gamma}_w = 250 \text{ s}^{-1}$, they seem to lean toward into areas greater than $8.5 \mu\text{m}$. Therefore, we confirm that the platelets are mainly concentrated in the CFL region close to the blood vessel wall at low wall shear rates, whereas they are more likely to accumulate in the NCFL region at high wall shear rates.

3.2. Effect of wall shear rate on platelet adhesion in blood flow

Owing to the behavior of margination, the platelets exhibit enhanced concentrations in the CFL region, which makes the platelets approach the wall surface and more likely form adhesion there. To model the adhesive ligands, we coated the solid tube wall with static DPD particles (as shown in yellow dots in Figure 1(b)), which can potentially bind to the receptors on the platelet membrane surface to form adhesive bonds. The effect of wall shear rate on platelet adhesion to the wall under four different shear rates is shown in Figure 4. The adhesion progress is shown with the relative time $\tau = t/T_0$. Under low shear rates, the adhesion process of platelets first shows progressive accumulation and then smoothly approaches saturation after $\tau = 0.5$. It is worth noting that the receptor-ligand binding interactions are hard to maintain under high shear rates;

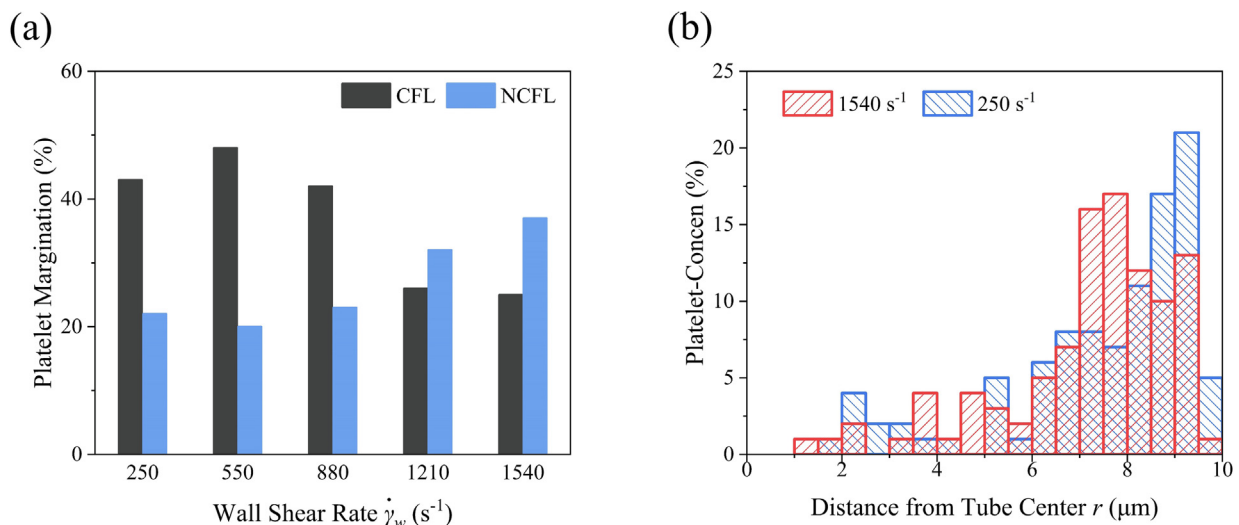


Figure 3. Dependence of platelet margination with respect to wall shear rate. (a) Percentage of flowing platelets in the CFL and the NCFL regions under different wall shear rates at $t = 0.27 \text{ s}$. (b) Concentration profiles of flowing platelets in the microtube at wall shear rate $\dot{\gamma}_w = 250 \text{ s}^{-1}$ and 1540 s^{-1} .

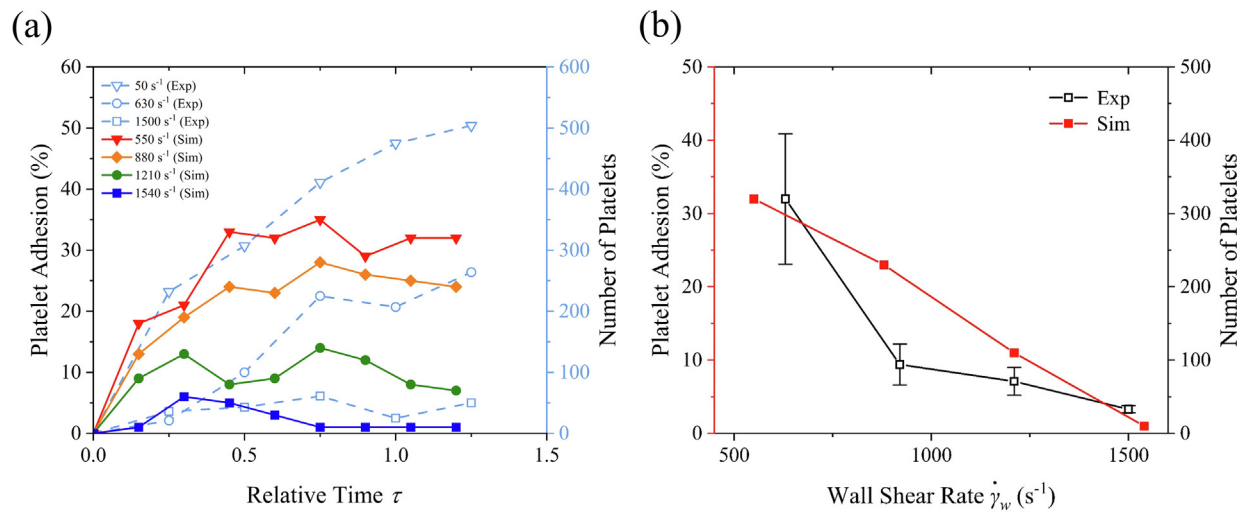


Figure 4. Effect of wall shear rate on platelet margination. (a) Temporal evolution of the platelet adhesion ratio at different shear rates. The relative time, $\tau = t/T_0$, is obtained through the physical time divided by a certain value at $T_0 = 0.15$ s. (b) Percentage of platelets adhered onto the FN-coated adhesion surface at $\dot{\gamma}_w = 550, 880, 1210$, and 1540 s^{-1} at $t = 0.18$ s ($\tau = 1.2$ in (a)). The experimental data are adapted from Ref.²⁴ with permission from Elsevier.

hence, it is difficult for the platelets to form firm adhesion to the wall. We then perform a quantitative analysis on the ratio of platelets adhered onto the FN-coated adhesion surface at $\tau = 1.2$, and the data shows that the maximum ratio of the platelet adhesion is 32% at $\dot{\gamma}_w = 550 \text{ s}^{-1}$, whereas it progressively decreases with increasing wall shear rates, as shown in Figure 4 (b). These results are consistent with previous experimental findings by Savage et al.²⁴ (as shown in Figure 4): On the one hand, the number of adhered platelets rapidly accumulates at the initial stage and then gradually reaches a plateau. On the other hand, a high shear rate is not conducive to the adhesive interaction between platelet and FN-coated adhesion surface, as shown by the fact that the number of adhered platelets decreases with increasing shear rate until the onset of no adhered platelets onto the FN-coated adhesion surface at about $\dot{\gamma}_w = 1540 \text{ s}^{-1}$. Meanwhile, there are also some differences between the simulation results and experimental data. For example, the time course of platelet adhesion in the experiment shows that the platelets continuously accumulated instead of reaching a plateau as revealed in our simulation. This can be explained by the fact that the platelets in the experimental system were constantly replenished, while only a finite number of platelets are placed in the simulation system. Actually, we adopt a periodic boundary condition along the flow direction, and the number of platelets placed inside the microtube is fixed. The choice of periodic boundary condition is inappropriate if there are dozens of platelets adhered onto the FN-coated tube wall surface, the current computational model cannot provide an accurate prediction on platelet margination ratio. Hence, the simulation results can only

provide a quasi-quantitative analysis on the platelet margination and adhesion behavior in microvascular blood flow. This deficiency could be addressed in future work by recourse to the general framework for open boundary systems for particle-based methods targeting simulations of blood flow,^{64,65} in which the model system includes an auxiliary buffer domain for blood cell generation.

3.3. Effect of fibrinogen concentration on platelet margination and adhesion in blood flow

It is known that the fibrinogen can mediate the adhesion of platelets to adhesion surface and to other platelets (aggregation).^{66–68} The fibrinogen concentration in pathological conditions (such as type 2 diabetes mellitus, T2DM) is higher than that in healthy ones. For example, it shows that the measured fibrinogen concentration level in T2DM patients is about 330–485 mg/dL, which is 2–4 times higher than that in healthy individuals (fibrinogen concentration level: 168–195 mg/dL).⁶⁹ Here, we perform detailed computational simulations of alterations to platelet adhesion and margination behaviors in response to the changes in fibrinogen concentration levels, by changing the number density of adhesion sites on the inner wall of the microtube. Specifically, we choose a low adhesion site density, N_1 , to represent the fibrinogen concentration level in normal condition. To reflect the changes in fibrinogen concentration levels based on the experimental data under pathological conditions, we consider five distinct adhesion site densities (i.e., N_2, N_3, N_4, N_5 , and N_6), and set their values to be $2N_1, 4N_1, 6N_1, 8N_1$, and $10N_1$, respectively.

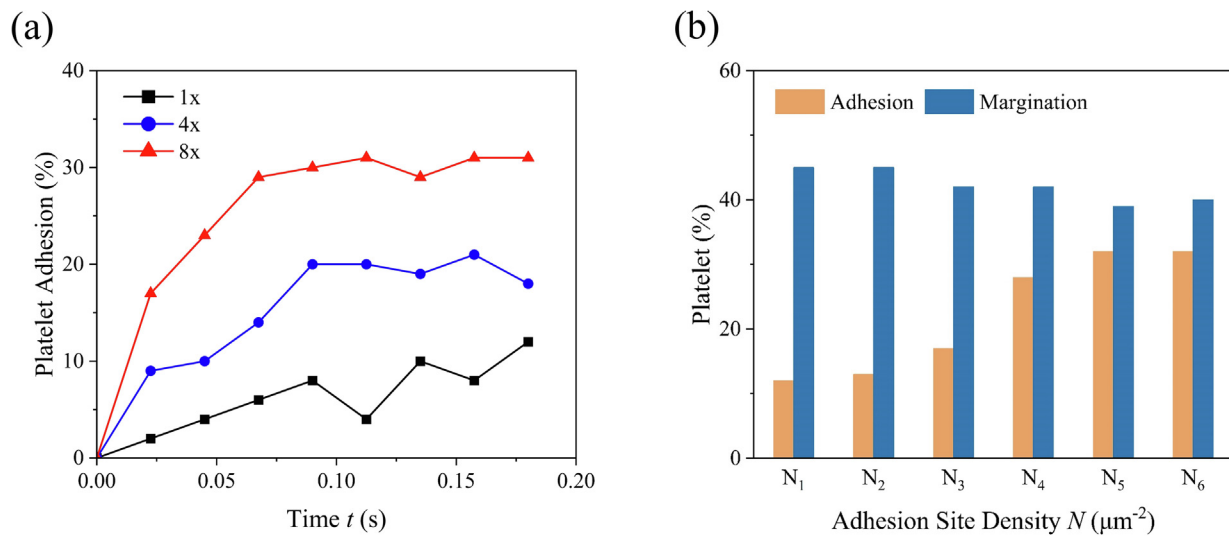


Figure 5. Effect of FN-coated adhesion site density on platelet adhesion and margination behaviors. (a) Time-dependent changes of platelet adhesion ratio at four different FN-coated adhesion site densities. (b) Percentage of adhered and margined platelets at different FN-coated adhesion site densities.

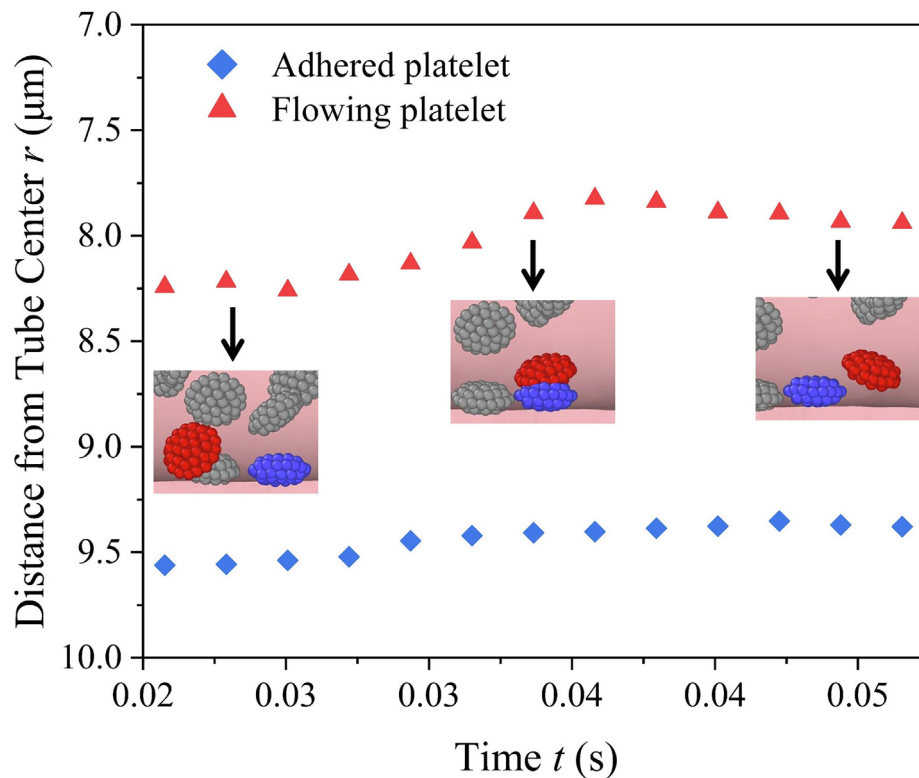


Figure 6. Effect of adhered platelets on the margination dynamics of individual flowing ones. The snapshots from left to right represent the dynamic process of the flowing platelet (highlighted in red color) sliding over the adhered one (highlighted in blue color) and moving away from the tube wall. The upper-triangle and diamond symbols represent the trajectories of the isolated flowing and adhered platelets when they first approach and then move apart from each other.

In this study, we choose the repulsive parameter between the platelets and the tube wall to be $a_{ij} = 2.5$, to prevent the platelets from penetrating solid

wall boundary. We employ the particle-based stochastic bond formation/dissociation model to capture the adhesive interactions between platelet

and FN-coated surface of the tube wall. For existing adhesive bonds, the adhesive force is defined by a spring force. In this study, we set the spring constant $k_s = 100$; hence, the adhesive force is much greater than the repulsive force to ensure a firm adhesion between platelet and FN-coated surface at low shear rates.

Figure 5(a) shows the time-dependent adhesion ratio of platelets on the FN-coated adhesion surface at four different adhesion site densities (i.e., N_1, N_3, N_5 , and N_6). A high fibrinogen concentration level results in a larger number of adhesive bonds connecting with the platelet membrane, causing strong adhesive interactions with the platelets. Therefore, the platelet adhesion ratio for N_5 (or N_6) shows a rapid increase initially, and maintains the prominent superiority in the subsequent process. Compared to the platelet adhesion ratio at N_1 , the ratio at N_5 (or N_6) increases significantly toward the steady stage, which is nearly three times higher than that at N_1 . The simulation results show that sufficient amounts of fibrinogen presented on the surface could mediate platelet adhesion under flow, which increased steadily with increasing fibrinogen concentration level, until reaching the maximum value. These findings are in agreement with previous experimental observations.^{70–72}

The platelet adhesion ratios and the corresponding margination ratios for different adhesion site densities at $t = 0.18$ s are shown in Figure 5(b). From a low value N_1 to a high value N_6 , the platelet adhesion ratio, as mentioned, increases first and then enters the plateau, whereas the platelet margination ratio drops slightly by about 8.9%. These results indicate that the adhered platelets have a considerable impact on the margination of the circulating platelets in microtube. For illustration and to help with the analysis, we track the motion of individual circulating platelets when they spot the adhered ones ahead of them, see Figure 6. When the circulating platelet (named PLT B, the one highlighted in red color in Figure 6) approaches the adhered platelet (named PLT A, the one highlighted in blue color in Figure 6), the hydrodynamic interaction induced by the adhered PLT A force the circulating PLT B to flip and move away from the tube wall, even far away from the CFL region. These results would provide some fundamental insights into how the adhered platelets influence the margination dynamics of the circulating ones.

It is known that the platelet margination and adhesion which are two critical and closely related steps in thrombus formation. Our simulation

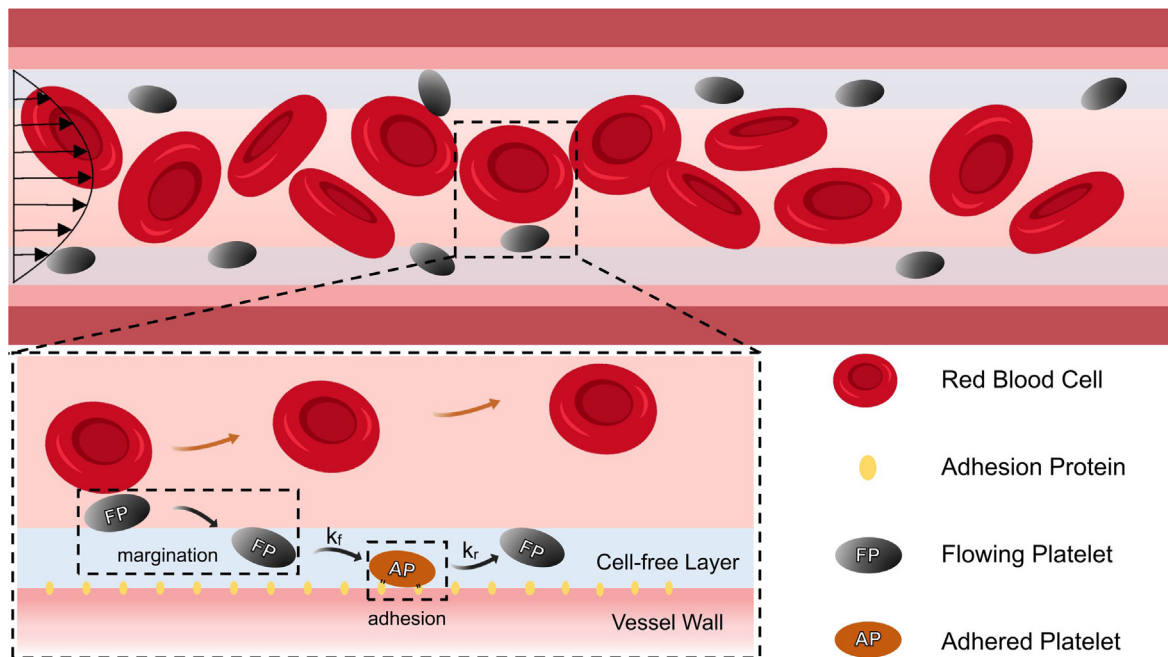


Figure 7. A mechanistic model of shear-induced margination and adhesion of platelets in microvascular blood flow. The occurrence of shear-induced RBC-platelet collisions cause the circulating platelets migrate toward the blood vessel wall through a process called margination. The platelets, which move along the vessel wall, can bind to immobilized fibrinogen through the stochastic kinetic adhesion interactions (k_f and k_r characterize the bond formation and dissociation rates) and deposit to vessel wall surface at low shear stresses. This may initiate the first stage of thrombus formation at the site of injured endothelium and triggers serious cardiovascular and microvascular diseases.

results show that the occurrence of shear-induced RBC-platelet collisions has a remarkable influence on the degree of platelet margination; meanwhile, the wall-induced lift force and the shear-induced displacement in wall-bounded flow cause the variation in near-wall platelet distribution. Therefore, when platelets enter the central region, they prefer to be pushed out to the CFL region, namely platelet margination. This could be helpful in understanding the movement and distribution of platelets under different physiological conditions.

In addition, the fibrinogen concentration level in pathological condition is much higher than that in healthy individuals, which enhance the plasma viscosity and mediate the platelet adhesion to surfaces coated with plasma proteins. Meanwhile, the platelets are more easily deposited at low shear rates, which in turn accelerates thrombus formation and triggers serious cardiovascular and microvascular diseases. Herein, by investigating the platelet adhesive dynamics under different flow conditions in the microtube with FN-coated wall surfaces, we find that the platelet adhesion is enhanced with the increase of FN concentration level but decreased with the increase of shear rate, which are consistent with available experimental results.^{24,70} Taken together, these findings provide useful insights into the platelet margination and adhesion dynamics, which may facilitate the understanding of the predominant processes governing the initial stage of thrombus formation. Based on these simulations findings, we developed a mechanistic model to show the shear-induced margination and adhesion of platelets in microvascular blood flow, see [Figure 7](#).

4. Conclusions

Platelets play an essential role in the process of hemostasis and thrombosis through their high-efficiency activation, adhesion, and aggregation. Nevertheless, their adverse reactions in blood vessels pose a serious threat to human health under pathological conditions including T2DM and sickle cell anemia, which have to be demonstrated to be extremely associated with the promotion of platelet hyperreactivity and responsiveness. Here, we employed a particle-based blood cell model to investigate the shear-induced margination and adhesion dynamics of platelets in microvascular blood flow, with a focus on the influences of wall shear rate, fibrinogen (FN) concentration level, and adhered platelets on the margination and adhesion processes of circulating ones. Without considering platelet adhesion, we found that the margination process of platelets is incredibly complicated and affected by multiple factors including blood hematocrit, cell-cell interactions, and wall shear rates. As shown from the computational simulations, the platelets tend to migrate rapidly toward the tube wall at a high

shear rate; interestingly, they tend to eject from the CFL region and aggregate into the NCFL region. Two competing underlying mechanisms can be used to understand such lateral transport processes of the circulating platelets. On the one hand, a high shear-induced collision frequency between the RBCs and platelets produces strong repulsion to the platelets, making them move far away from the RBCs and migrate gradually to tube walls. On the other hand, the drastic flow fluctuations provide a centripetal lift force for platelets' movements. Subsequently, we probe the effect of wall shear rate on the platelet adhesion in the microtube. We found that the number of adhered platelets rapidly accumulates at the initial stage and then gradually reaches a plateau, and the platelet adhesion ratio decreased with increasing wall shear rate, which are consistent with available experimental measurements.²⁴ Finally, we simulate the platelet adhesion dynamics under different wall shear rates by conducting DPD simulations of microvascular blood flow with FN-coated adhesion surface. We showed that an elevated FN concentration level can cause enhanced platelet adhesion under pathological conditions. We also found that the adhered platelets has an adverse effect on the circulating platelet margination.

In this study, our main effort in the current study is to investigate the combined effects of platelet margination with adhesion. We note that the platelet adhesion dynamics is a protein-dependent process, and different receptor-ligand binding exhibits different adhesive characteristics. Even for the same or similar receptor-ligand bonds, the bond formation and dissociation kinetics may be regulated by several biophysical factors such as force history, contact duration, and approach velocity.^{73–75} In future study, we will perform further sensitivity studies as we refine our platelet adhesive dynamics model, by learning the adhesive parameters between platelets and adhesive proteins using machine learning tools as well as microfluidic experiments that will give us big data sets for both trajectories and velocities. Such computational simulation results and their comparisons with available experimental data would be helpful in understanding the predominant processes governing the initial stage of thrombus formation.

CRedit authorship contribution statement

Lujuan Li: Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing, Visualization. **Shuo Wang:** Formal analysis, Investigation, Methodology, Writing – review & editing, Visualization. **Keqin Han:** Formal analysis, Methodology, Writing – review & editing. **Xiaoqing Qi:** Formal analysis, Methodology, Writing –

review & editing. **Shuhao Ma:** Formal analysis, Methodology, Writing – review & editing. **Li Li:** Formal analysis, Writing – review & editing. **Jun Yin:** Formal analysis, Writing – review & editing. **Dechang Li:** Conceptualization, Supervision, Writing – review & editing. **Xuejin Li:** Conceptualization, Formal analysis, Supervision, Writing – original draft, Writing – review & editing. **Jin Qian:** Conceptualization, Formal analysis, Supervision, Writing – original draft, Writing – review & editing.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary Data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jmb.2022.167824>.

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